

DBMS TASK – SELLER AND INVENTORY MANAGEMENT

Objective

To create Seller and Inventory tables, establish relationships between sellers, products and stock, maintain seller product information, track product availability, and generate inventory status reports.

1. DATABASE DESIGN

1.1 SELLER TABLE

Attribute	Data Type	Key / Constraint
Seller_ID	NUMBER(5)	Primary Key
Seller_Name	VARCHAR2(100)	NOT NULL
Seller_Email	VARCHAR2(100)	UNIQUE
Contact_Number	VARCHAR2(15)	UNIQUE
Business_Type	VARCHAR2(50)	NOT NULL

1.2 PRODUCT TABLE

Attribute	Data Type	Key / Constraint
Product_ID	NUMBER(5)	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Category	VARCHAR2(50)	NOT NULL
Price	NUMBER(10,2)	NOT NULL
Stock	NUMBER(5)	NOT NULL

1.3 INVENTORY TABLE

Attribute	Data Type	Key / Constraint
Inventory_ID	NUMBER(5)	Primary Key
Product_ID	NUMBER(5)	Foreign Key
Seller_ID	NUMBER(5)	Foreign Key

Stock_Quantity	NUMBER(5)	NOT NULL
Stock_Status	VARCHAR2(20)	NOT NULL
Last_Updated	DATE	NOT NULL

2. TABLE RELATIONSHIP

The relationship between the **Seller**, and **Inventory** tables is established using Primary Keys and Foreign Keys.

1. Create Seller and Inventory Tables

Seller Table

The Seller table stores the details of sellers.

```
SQL> CREATE TABLE Seller (
2     Seller_ID NUMBER PRIMARY KEY,
3     Seller_Name VARCHAR2(50),
4     Contact VARCHAR2(15),
5     Email VARCHAR2(50)
6 );
```

Table created.

Inventory Table

The Inventory table stores product stock details supplied by sellers.

```
SQL> CREATE TABLE Inventory (
2     Inventory_ID NUMBER PRIMARY KEY,
3     Product_ID NUMBER,
4     Seller_ID NUMBER,
5     Stock_Quantity NUMBER,
6     Stock_Status VARCHAR2(20)
7 );
```

Table created.

Insert Seller Records

```
SQL> INSERT INTO Seller VALUES
  2 (101, 'Arun Traders', '9876543210', 'arun@gmail.com');

1 row created.

SQL> INSERT INTO Seller VALUES
  2 (102, 'Priya Stores', '9876501234', 'priya@gmail.com');

1 row created.

SQL> INSERT INTO Seller VALUES
  2 (103, 'Kumar Enterprises', '9845671230', 'kumar@gmail.com');

1 row created.

SQL> INSERT INTO Seller VALUES
  2 (104, 'Sri Lakshmi Traders', '9988776655', 'lakshmi@gmail.com');

1 row created.

SQL> INSERT INTO Seller VALUES
  2 (105, 'Green Mart', '9123456780', 'greenmart@gmail.com');

1 row created.
```

Insert Inventory Records

```
SQL> INSERT INTO Inventory VALUES
  2 (1, 1, 101, 50, 'Available');

1 row created.

SQL> INSERT INTO Inventory VALUES
  2 (2, 2, 102, 30, 'Available');

1 row created.

SQL> INSERT INTO Inventory VALUES
  2 (3, 3, 103, 0, 'Unavailable');

1 row created.

SQL> INSERT INTO Inventory VALUES
  2 (4, 4, 104, 25, 'Available');

1 row created.

SQL> INSERT INTO Inventory VALUES
  2 (5, 5, 105, 0, 'Unavailable');

1 row created.
```

Display Seller Records

```
SQL> SELECT * FROM Seller;
```

SELLER_ID	SELLER_NAME	CONTACT
101	Arun Traders	9876543210
102	Priya Stores	9876501234
103	Kumar Enterprises	9845671230
104	Sri Lakshmi Traders	9988776655
105	Green Mart	9123456780

Display Inventory Records

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS
1	1	101	50	Available
2	2	102	30	Available
3	3	103	0	Unavailable
4	4	104	25	Available
5	5	105	0	Unavailable

3. Establish Relationships Between Sellers, Products, and Stock

The relationship between Seller, Product, and Inventory tables is established using foreign keys.

- Seller_ID is the Primary Key in the Seller table.
- Product_ID refers to the Product table.
- Seller_ID in Inventory refers to the Seller table.
- Inventory connects sellers with products and maintains stock information.

```
SQL> SELECT S.Seller_ID,
2         S.Seller_Name,
3         I.Product_ID,
4         I.Stock_Quantity,
5         I.Stock_Status
6 FROM Seller S
7 JOIN Inventory I
8 ON S.Seller_ID = I.Seller_ID;
```

SELLER_ID	SELLER_NAME	PRODUCT_ID
101	Arun Traders	1
50	Available	
102	Priya Stores	2
30	Available	
103	Kumar Enterprises	3
0	Unavailable	
104	Sri Lakshmi Traders	4
25	Available	
105	Green Mart	5
0	Unavailable	

4. Maintain Seller Product Information

Seller product information can be maintained using UPDATE and displayed using an INNER JOIN.

Update Seller Information

```
SQL> UPDATE Seller
2 SET Contact = '9876543211'
3 WHERE Seller_ID = 101;
```

1 row updated.

Display Seller Product Information Using INNER JOIN

```
SQL> SELECT S.Seller_ID,
2         S.Seller_Name,
3         I.Product_ID,
4         I.Stock_Quantity,
5         I.Stock_Status
6 FROM Seller S
7 INNER JOIN Inventory I
8 ON S.Seller_ID = I.Seller_ID;
```

SELLER_ID

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS

```
-----
101
ABC Traders
1001          50 Available

102
XYZ Store
1002          30 Available
```

SELLER_ID

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS

```
-----
103
Kumar Products
1003          15 Available

104
Sri Enterprises
```

SELLER_ID

SELLER_NAME

104
Sri Enterprises

SELLER_ID

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS

```
-----
1004          0 Out of Stock

105
Green Mart
1005          40 Available
```

5. Track Available and Unavailable Products

Display Available Products

```
SQL> SELECT Product_ID, Stock_Quantity, Stock_Status
2  FROM Inventory
3  WHERE Stock_Status = 'Available';
```

PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
1	50	Available
2	30	Available
4	25	Available

Display Unavailable Products

```
SQL> SELECT Product_ID, Stock_Quantity, Stock_Status
2  FROM Inventory
3  WHERE Stock_Status = 'Unavailable';
```

PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
3	0	Unavailable
5	0	Unavailable

Update Stock Status

```
SQL> UPDATE Inventory
2  SET Stock_Status = 'Available'
3  WHERE Stock_Quantity > 0;
```

4 rows updated.

```
SQL> UPDATE Inventory
2  SET Stock_Status = 'Unavailable'
3  WHERE Stock_Quantity = 0;
```

1 row updated.

6. Generate Inventory Status Reports

Complete Inventory Status Report

```
SQL> SELECT S.Seller_ID,
2         S.Seller_Name,
3         I.Product_ID,
4         I.Stock_Quantity,
5         I.Stock_Status
6 FROM Seller S
7 INNER JOIN Inventory I
8 ON S.Seller_ID = I.Seller_ID
9 ORDER BY I.Stock_Status;
```

```
SELLER_ID
```

```
SELLER_NAME
```

```
PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
```

```
101
ABC Traders
1001          50 Available
```

```
102
XYZ Store
1002          30 Available
```

```
SELLER_ID
```

```
SELLER_NAME
```

```
PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
```

```
105
Green Mart
```

```
105
Green Mart
1005          40 Available
```

```
103
Kumar Products
```

```
SELLER_ID
```

```
SELLER_NAME
```

```
PRODUCT_ID STOCK_QUANTITY STOCK_STATUS
```

```
1003          15 Available
```

```
104
Sri Enterprises
1004          0 Unavailable
```

Available Product Report

```
SQL> SELECT S.Seller_Name,
2         I.Product_ID,
3         I.Stock_Quantity,
4         I.Stock_Status
5 FROM Inventory I
6 INNER JOIN Seller S
7 ON I.Seller_ID = S.Seller_ID
8 WHERE I.Stock_Status = 'Available';
```

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS

ABC Traders
1001 50 Available

XYZ Store
1002 30 Available

Kumar Products
1003 15 Available

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS

Green Mart
1005 40 Available

Unavailable Product Report

```
SQL> SELECT S.Seller_Name,
2         I.Product_ID,
3         I.Stock_Quantity,
4         I.Stock_Status
5 FROM Inventory I
6 INNER JOIN Seller S
7 ON I.Seller_ID = S.Seller_ID
8 WHERE I.Stock_Status = 'Unavailable';
```

SELLER_NAME

PRODUCT_ID STOCK_QUANTITY STOCK_STATUS

Sri Enterprises
1004 0 Unavailable

Conclusion

The Seller and Inventory tables were successfully created and populated with records. Relationships were established between sellers, products, and inventory using primary and foreign keys. Seller product information was maintained using UPDATE and INNER JOIN queries. Available and unavailable products were tracked, and inventory status reports were generated successfully.